

महाराष्ट्र वन सेवा मुख्य परीक्षा - 2025
दि. 07 मे, 2026



2025

Q24

BOOKLET NO.

109138

Forest Services
Statistics

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – Candidates will be allowed to use Scientific (Non-programmable type) calculators.

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P.T.O.



SECTION – A

- Q1.** Solve **any five** out of seven. **(8×5=40)**
- Q1.** (a) There are two types of machines, R and S, used in agriculture. These machines are to be sold in the market. A sales executive is looking after sales of these machines. The probability that executive sells machine R is 0.55 and the probability that he sells machine S is 0.45. Assuming that the two machines are sold independently what is probability that **8**
- i) at least one machine is sold.
 - ii) machine R is sold given that executive has sold at least one machine.
- Q1.** (b) Define : **8**
- i) Parameter.
 - ii) Statistic.
 - iii) Estimator.
 - iv) Unbiased estimator.
- Q1.** (c) If $(x_i, y_i) i = 1, 2, \dots, n$ is bivariate data on (X, Y) , fit the line of regression of y on x , $y = a + bx$ by using the least squares method. **8**
- Q1.** (d) Explain any five principal steps in a sample survey. Also explain sampling and non-sampling errors. **8**
- Q1.** (e) Define : **8**
- i) Hypothesis.
 - ii) Null hypothesis.
 - iii) Composite hypothesis.
 - iv) Error and its types in testing of hypothesis.
- Q1.** (f) If $(X, Y) \sim B \vee N (\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ then state the p.d.f. of (X, Y) . If $\rho = 0$ then show that X and Y are independent. State the distributions of X and Y . **8**
- Q1.** (g) Explain simple random sampling and stratified random sampling. Also explain when do you prefer stratified random sampling over simple random sampling by giving an example. **8**
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- Q2. (a) A box contains 6 black and 4 blue marbles. Three marbles are drawn together at random from the box. If X denotes the number of black marbles drawn then 15
- State all possible values of X .
 - Is X a continuous variable or discrete variable ?
 - If $Y = \frac{6}{1+X}$, find $E(Y)$.
- Q2. (b) State and prove Cramer-Rao inequality for the variance of unbiased estimator of function of parameter. Also state the regulating conditions for this inequality. 15
- Q2. (c) If the two regression equation are $5x - 4y + 25 = 0$ and $3x - 5y + 33 = 0$, identify the line of regression of y on x and of x on y . Give reason. Also find : 10
- Correlation coefficient between x and y .
 - Value of x when $y = 5$.
 - Value of y when $x = 4$.

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- Q3. (a) State and prove Gauss-Markov theorem for the model $Y = X\beta + e$ where Y is $n \times 1$ vector of observations, X is $n \times p$ matrix of levels of regressor variables, β is $p \times 1$ vector of regression coefficients, and e is $n \times 1$ vector of errors, $E(e) = 0$ and $\text{Var}(e) = \sigma^2 I$. 15

- Q3. (b) Fertilizers F_1, F_2, F_3 and F_4 were used in an agricultural experiment. The yields from plots (in kg) are given below : 15

F_3 10	F_1 14	F_2 16	F_4 13	F_2 14
F_2 12	F_4 11	F_3 12	F_1 16	F_1 12
F_4 10	F_2 12	F_1 11	F_3 11	F_4 13
F_2 13	F_3 15	F_4 13	F_2 12	F_1 11

Identify the design. Analyze the data by stating the model used, hypothesis to be tested. Also give the estimate of the error variance.

(Given : $F_{3,16,0.05} = 3.24$, $F_{4,20,0.05} = 2.87$, $F_{16,3,0.05} = 8.70$)



- Q3. (c)** The length of groundnut pod follows normal distribution with mean 3 cm and variance 4. Then answer the following : 10

- i) What is the probability that a randomly selected pod will have length below 4 cm ?
- ii) What is the median length of the groundnut pod ?
- iii) If 10 pods are selected at random then how many pods do you expect to have length between 2 cm and 4 cm ?

(Given : If Z is ZNV then $P(Z \leq 0.5) = 0.6915$)

- Q4. (a)** A random sample of size 16 is drawn from a normal population with mean μ and variance 25. If the critical region for testing $H_0 : \mu = 4$ against $H_1 : \mu = 6$ is $\sum_{i=1}^{16} x_i \geq 70$, where x_i is i^{th} unit in the sample, find : 15

- i) $P(\text{Type I error})$
- ii) $P(\text{Type II error})$
- iii) Power of the test.

(Given : If Z is SNV, then $P(Z < 0.3) = 0.6179$, $P(Z < 0.5) = 0.6915$, $P(Z < 1.3) = 0.9032$.)

- Q4. (b)** What are factorial experiments ? 15

State any three advantages of factorial experiments over simple experiments.

In 2^2 factorial design with factors A and B show that interaction AB is same as interaction BA, by assuming that design is conducted in RBD with r blocks.

For the above factorial design give a blank ANOVA table by stating hypothesis to be tested and various formulae used.

- Q4. (c)** Under the conditions to be stated show that Poisson distribution is a limiting case of the Binomial distribution. 10
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**SECTION – B**

- Q5.** Solve **any five** out of seven. **(8×5=40)**
- Q5.** (a) Define : **8**
- i) Producers' Risk.
 - ii) Consumers' Risk.
 - iii) Process Average fraction defective.
- Q5.** (b) Explain the significance of Artificial variables in simplex method of L.P.P. Also describe briefly the two phase method of solving L.P. Problem with Artificial variables. **8**
- Q5.** (c) Explain Box – Jenkins ARIMA modelling outlining its key steps. **8**
- Q5.** (d) Write short notes on the following terms : **8**
- i) z (or σ) scores
 - ii) Normalised scores
 - iii) Percentile scores.
- Q5.** (e) Derive the optimal economic lot size formula for an inventory model with uniform demand, infinite production and shortages are not allowed. **8**
- Q5.** (f) Write a short notes on : **8**
- i) Cost of living (consumer price) index number.
 - ii) Index of Industrial production.
- Q5.** (g) Describe : **8**
- i) CRUDE DEATH RATE.
 - ii) Specific Death Rates.
 - iii) Mortality table or Life table.



- Q6. (a) (I)** If t is a lifetime random variable with hazard function $h(t)$ and mean residual lifetime $r(t)$ at age ' t ', then show that

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i) $h(t) = \frac{r'(t) + 1}{r(t)}$

ii) $r(t) = \int_0^{\infty} e^{H(t) - H(t+u)} du.$

- (II) Obtain hazard function of the lifetime random variable having density function

$$f(t) = (\alpha + \beta t) e^{-(\alpha t + \frac{1}{2}\beta t^2)}, t > 0, \alpha > 0, \beta > 0$$

= 0, otherwise

Also identify the nature of hazard function.

- Q6. (b) i)** The Bombay Transport Company has trucks available at four different sites in the given following table :

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Site :	A	B	C	D
No. of Trucks :	5	10	7	3

Customer W, X and Y require trucks as shown below :

Customer :	W	X	Y
No. of Trucks :	5	8	10

Variable costs of getting trucks to the customers are :

From A to W → Rs. 7, to X → Rs. 3, to Y → Rs. 6.

From B to W → Rs. 4, to X → Rs. 6, to Y → Rs. 8.

From C to W → Rs. 5, to X → Rs. 8, to Y → Rs. 4.

From D to W → Rs. 8, to X → Rs. 4, to Y → Rs. 3.

Solve the above transportation problem.

- ii) Consider a 'modified' form of 'matching biased coins' game problem. The matching player is paid Rs. 8 if the two coins turn both heads and Re. 1 if the coins turn both tails. The non-matching player is paid Rs. 3 when the two coins do not match. Given the choice of being the matching or non-matching player, which one would you choose and what would be your strategy ?



- Q6.** (c) i) Explain different methods of collection of official statistics. **10**
 ii) Define Fisher's index number and show that it satisfies Factor Reversal test.
 iii) Define Linear model. Write down the generalised linear model. Prove that $\underline{e} \sim N_m(0, \sigma^2 I_{m \times n})$.

- Q7.** (a) From the following data calculate : **15**
 i) Laspeyre's Price index for 2005 using 1995 as the base.
 ii) Marshall – Edgeworth Price index number with 1995 as the base and 2005 as current year.
 iii) Bowley's Price index number with base year as 1995 and current year as 2005.
 iv) Fisher's quantity index number with base 2005 and current year as 1995.
 v) Walsch Price index taking current year as 2005 and base year 1995.

Commodities	1995		2005	
	Price	Quantity	Price	Quantity
A	20	8	40	6
B	50	10	60	5
C	40	15	50	15
D	20	20	20	25

- Q7.** (b) Given the following table for l_x , the number of rabbits living at age x, complete the life table for rabbits. **15**

x :	0	1	2	3	4	5	6
l_x :	100	90	80	75	60	30	0

X, Y, Z are rabbits of age 1, 2 and 3 years respectively. Find the probability that at least one of them will be alive for one year more.

- Q7.** (c) Attempt **any two** from the following : **10**
 i) Explain clearly the basis of and working of control charts for mean and range. State the control limits of $\bar{X}$ and R charts.
 ii) When to use S chart ? How to estimate the standard deviation σ ? Give control limits for S chart.
 iii) Derive the control limits P chart, nP chart and c chart.



- Q8. (a)** In machine maintenance a mechanic repairs four machines. The meantime between service requirement is 5 hours for each machine and forms an exponential distribution. The mean repair time is one hour and also follows the same distribution pattern machine down time costs Rs. 25 per hour and machine costs Rs. 55/day for 8 hours. **15**
- i) Find the expected number of operating machines.
 - ii) Determine the expected down time cost per day.
 - iii) Would it be economical to engage two mechanics each repairing only two machines ?

- Q8. (b)** Complete i) G.F.R. ii) S.F.R. iii) T.F.R. and iv) the gross production rate from the data given below : **15**

Age group of child bearing females	Number of women('000)	Total births
15 – 19	16.0	260
20 – 24	16.4	2244
25 – 29	15.8	1894
30 – 34	15.2	1320
35 – 39	14.8	916
40 – 44	15.0	280
45 – 49	14.5	145

Assume that the proportion of female births is 46.2 percent.

- Q8. (c)** Attempt **any two** from the following : **10**
- i) Explain in detail single sampling plan and hence state the formulae for producers' risk and consumers' risk using Hyper geometric distribution and Binomial distribution.
Also explain probability of accepting a lot of quality 'P'.
 - ii) Write short notes on double sampling plan.
 - iii) Define survival function, Hazard function.
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